

Joaquim Ortiz-Haro

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Citizenship: Spanish

I am a Research Scientist at Helsing AI (formerly Keybotic) in Barcelona, where I work on deep reinforcement learning for robotic locomotion. Previously, I was a Postdoctoral Researcher in Robotics at New York University (NYU) in the Machines in Motion Laboratory, supervised by Prof. Ludovic Righetti. My research interests include locomotion, task and motion planning, control, and deep learning for robotics. I received my PhD in Robotics from TU Berlin, supervised by Prof. Marc Toussaint.

EDUCATION

- **PhD in Robotics** Sep. 2019 - Jan. 2024
TU Berlin (Germany) and IMPRS-IS (Max Planck Research School). Focus on task and motion planning, nonlinear optimization, and deep learning for robotics. Advisor: Marc Toussaint.
- **Master of Mathematics** 2017 - 2019
UPC (Polytechnic University of Catalonia, Barcelona), Faculty FME. Average qualification: 9.38/10. Exchange student at LAAS-CNRS (French National Research Center).
- **Bachelor of Mechanical Engineering** 2013 - 2017
UPC (Polytechnic University of Catalonia, Barcelona), Faculty ETSEIB. Top Class Honours. Ranked 2nd out of 350 graduate students. Average qualification: 8.84/10. Exchange student at ETH Zurich (Switzerland).
- **High School** 2014 - 2017
Honour Mention in the Spanish University Entrance Exam (13.36/14). Scholarship to study the first year at university. Gold Medal in the XXVI Chemistry Spanish National Olympiad.

WORK EXPERIENCE

- **Robotics and AI Research Scientist – Helsing AI/Keybotic** Jan. 2025 - Ongoing
Main research project: state-of-the-art vision-based locomotion for legged robots. Trained RL policies, developed new architectures and training pipelines, conducted sim-to-sim evaluation and sim-to-real transfer, and deployed policies on real robots.
- **Postdoctoral Researcher, Robotics, NYU** Feb. 2024 - Jan. 2025
Machines in Motion Laboratory, New York University (NYU), under the direction of Professor Ludovic Righetti. Conducting research in task and motion planning, control, and deep learning for robotics. Leading research projects, supervising PhD and Master's students, and collaborating with team members on various research projects.
- **PhD Student, Robotics** Sep. 2019 - Jan. 2024
See section Education.
- **Research Intern, Machine Learning** 2019
Worked on optimal transport and theoretical machine learning. Artificial Intelligence and Machine Learning Group at UPF (University Pompeu Fabra, Barcelona). Supervisor: Gergely Neu.
- **Research Intern, Robotics** 2018
Conducted research on contact and trajectory optimization using an augmented Lagrangian algorithm. Humanoid Robotics at LAAS-CNRS (French National Research Center). Supervisor: Nicolas Mansard.
- **Intern, Robotics Engineer** 2018
Performed geometric and semantic analysis of 3D point clouds. Scaled Robotics (start-up, Barcelona).
- **Seat Autonomous Car Competition** 2017
Member of the IRI (Robotics Institute, Barcelona) team for an autonomous driving competition, scale 1:10.

- **Research Assistant, SLAM** 2017
Developed parallel tracking and mapping algorithms for an event-based camera. IRI (Robotics Institute, Barcelona). Supervisor: Joan Solà.
- **Teaching Assistant** 2015 - 2017
Taught mathematics to first-year engineering students at ETSEIB, UPC.
- **Private Academy Teacher** 2014
Taught university-level engineering lessons at a private academy.

PROGRAMMING

- **C++:** Main tool for robotics research during PhD and Postdoc. Utilized in internships at robotics companies and robotics competitions. Proficient with BOOST, ROS, PCL, OpenCV, EIGEN, OMPL, and developed custom C++ libraries with Python bindings.
- **Python:** Used for deep learning research during PhD (PyTorch). Applied in master's thesis for robotics and numerical optimization (SciPy, NumPy, etc.) and various programming courses.
- **Matlab:** Utilized in bachelor thesis (robotics and computer vision) and university courses.
- **Other:** Experienced with Arduino, PlatformIO, ANSYS, SolidWorks, Minitab, and Autodesk Fusion.

LANGUAGES

- **Spanish and Catalan:** Native.
- **English:** Professional proficiency. Scientific publications, bachelors and masters theses written in English. Advanced Certificate (CAE) from Cambridge University, C1.2 (2013).
- **French:** Official certificate DELF B2 (2019). Six-month exchange in Toulouse.
- **German:** Official certificate B2.2 from Goethe Institute (2017). Engineering courses conducted in German at ETH. Attended two summer language courses in Germany.

PERSONAL SKILLS

Passionate about research, problem-solving, and learning. Experienced in interdisciplinary and international teams. Dedicated, honest, and hardworking. I love teaching and sharing my knowledge. In my free time, I enjoy climbing, cycling, and reading books. I also play the guitar and am learning the piano.

Last updated: March 2026.